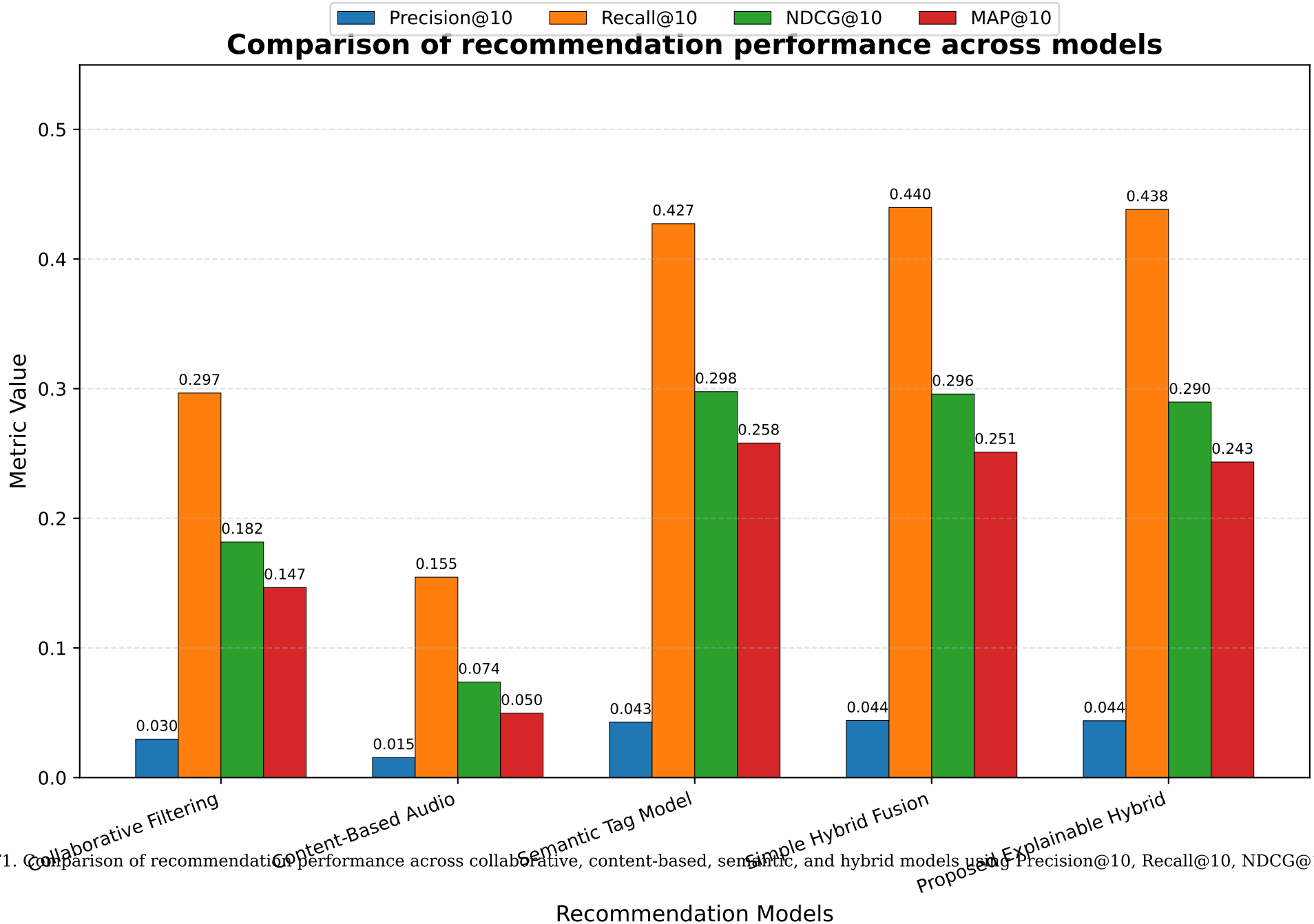

Figure RQ1-F1. Comparison of recommendation performance across collaborative, content-based, semantic, and hybrid models using Precision@10, Recall@10, NDCG@10, and MAP@10.